

SPILLTRACE

Maritime Pollution Attribution System

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How to Read This PRD

This version is intentionally written for a team that is about to implement the project. It first explains every important term in very simple language, then tells exactly where the data for that term/step will come from, and only then gives the implementation plan in order.

The internal SIH deck is the product baseline: detect oil-like slicks with Sentinel-1 SAR, verify look-alikes, reverse-drift to an origin region, reconstruct AIS trajectories, and rank vessels with explainable scoring. The deck also states that attribution is probability/investigative evidence, not automatic legal proof.

PART A — FIRST UNDERSTAND THE TERMS

Do not start coding before the team understands these words.

Term	Simple Meaning	Why We Need It
Oil Spill	Sea mein accidentally/illegally release hua oil jo water surface par spread hota hai.	We want to detect and trace it.
Satellite	Space mein ghoomta sensor jo Earth ki images/data collect karta hai.	Source of remote observations.
Sentinel-1	Copernicus ka radar satellite mission jo Earth ko SAR se observe karta hai.	Main satellite source.
SAR	Synthetic Aperture Radar — camera jaisa nahi; radar signal bhejkar surface ka radar response measure karta hai.	Oil-like dark/low-backscatter regions detect karne ke liye.
SAR Image	SAR se aayi 2D raster image jisme har pixel radar measurement represent karta hai.	U-Net ka input.
GRD	Ground Range Detected — Sentinel-1 ka processed product, MVP ke liye comparatively easier starting point.	Hum initial Sentinel-1 data ke liye GRD use karenge.
VV / VH	SAR ki do common polarization channels; dataset mein ye do channels image ke do inputs ho sakte hain.	ML input features.
Raster	Grid of pixels — image ko computer mein store karne ka format.	Satellite data.
GeoTIFF	Geographic information ke saath raster/image file.	Processed satellite output storage.
U-Net	Deep-learning segmentation model jo image ke pixels ko classify karke oil ka mask banata hai.	Oil spill detection.
Segmentation	Image ke har relevant pixel ko label karna; yahan oil vs non-oil.	Spill boundary nikalna.
Mask	Black/white ya probability image jahan oil pixels marked hote hain.	U-Net output.
Polygon	Map par oil spill ki boundary ka shape.	Dashboard, GIS and drift input.
Look-alike	Aisa dark SAR pattern jo oil jaisa dikhe par oil na ho, e.g. some natural/wake/low-wind effects.	False-positive reduction.
Wind	Hawa ki direction aur speed.	Oil movement and verification.
Ocean Current	Samundar ke paani ka direction/speed.	Oil movement/drift.
OpenDrift	Open-source particle drift modelling framework.	Oil ko environmental forcing ke saath simulate karna.
OpenOil	OpenDrift ke andar oil-specific model.	Oil-specific drift/weathering simulation.
Hindcast / Reverse Drift	Observed spill se time ko peeche le jaakar possible origin find karna.	Likely discharge region.
Particle	Simulation mein oil ko represent karne wala virtual point.	Drift calculation.

Probability Region	Ek area jahan origin hone ki likelihood comparatively high hai; exact point nahi.	Attribution input.
AIS	Automatic Identification System — ships ka broadcast tracking system.	Ship identity/location/movement.
MMSI	AIS mein vessel/station ka unique maritime identifier.	Vessel grouping.
SOG	Speed Over Ground — ship ki actual ground speed.	Scoring/trajectory.
COG	Course Over Ground — ship kis direction mein move kar raha hai.	Scoring/trajectory.
AIS Trajectory	Time-ordered AIS positions ko connect karke ship ka historical route.	Origin matching.
PostgreSQL	Main relational database.	Cases, vessels, AIS metadata, scores.
PostGIS	PostgreSQL ka geospatial extension.	Points, lines, polygons and spatial queries.
Redis	Fast memory store; hum cache aur background-job queue ke liye use karenge.	Jobs/cache.
FastAPI	Python framework jisse backend APIs banayenge.	Frontend ↔ processing/database connection.
Next.js	React-based web framework jisse dashboard banayenge.	Frontend.
TypeScript	JavaScript ka typed version.	Frontend code safety.
MapLibre	Interactive web maps banane ka mapping library.	Spill/vessel/drift map.
GDAL/Rasterio	Geospatial raster processing tools/libraries.	SAR preprocessing and raster operations.
Object Storage	Large files rakhne ki storage, e.g. S3/MinIO.	GeoTIFF, NetCDF, reports.
Worker	Background process jo heavy task karta hai.	ML, drift, downloads.
API	Ek software interface jisse frontend backend se data/request exchange karta hai.	System integration.
Docker	Application aur dependencies ko container mein package/run karne ka tool.	Same environment for team/deployment.

PART B — EXACTLY DATA KAHA SE AAYEGA?

There are two kinds of data: (1) training data used to teach the ML model and (2) runtime/live data used when an analyst investigates a new spill.

Data	Source	How We Get It	Use
Sentinel-1 satellite imagery	Copernicus Data Space	AOI + acquisition date/time → Sentinel-1 GRD product	Live/investigation data
Oil/no-oil/look-alike training images	Public research dataset on Zenodo	Download labelled VV/VH images + masks	Training/validation/test
Wind	Copernicus Marine wind products; operational atmospheric provider can be added	Time + spatial subset	Verification + drift
Ocean current	Copernicus Marine	Time + spatial subset of current fields	Drift
AIS prototype stream	AISStream	Backend subscribes to geographic bounding box/message types	Runtime vessel positions
AIS production	Authorized/contracted/government source	Provider-specific API/feed	Production
Vessel metadata	AIS static/voyage messages and legally available registry/provider data	MMSI/IMO/name/type etc.	Vessel profile
Model output	Our U-Net	New Sentinel-1 image → mask	Runtime derived data
Drift output	Our OpenOil/OpenDrift run	Spill + wind + current → particles/origin	Runtime derived data
Scores	Our scoring engine	Origin + AIS + trajectory features	Runtime derived data

Current Source Notes

- Copernicus Data Space currently lists Sentinel-1 GRD worldwide from October 2014 onward and documents catalogue/API access. [cite](#)[turn0search1](#)[turn0search0](#)
- Sentinel-1 GRD documentation lists VV/VH/HH/HV bands and explains that GRD is detected, multi-looked and projected to ground range. [cite](#)[turn0search3](#)

- The public Zenodo training/validation dataset includes 685 no-oil and 685 look-alike images with corresponding masks; its images are 2048×2048×2 VV/VH. [\[cite\]turn0search4](#)
- The public Zenodo test set contains 150 oil, 150 look-alike and 150 no-oil images with corresponding ground-truth masks. [\[cite\]turn0search2](#)
- Copernicus Marine provides programmatic access to ocean/wind datasets through its toolbox/API options. [\[cite\]turn0search6](#)
- OpenDrift's official docs expose OpenOil as the oil-drift model and describe environmental forcing such as wind/current. [\[cite\]turn0search5](#)[\[turn0search8\]](#)
- AISStream currently provides a WebSocket AIS stream with geographic bounding-box filters and requires the API key to remain server-side; direct browser connections are not permitted. [\[cite\]turn1search0](#)[\[turn1search2\]](#)

PART C — OUR COMPLETE SYSTEM IN ONE PICTURE

USER → AOI/DATE → SENTINEL-1 → PREPROCESS → U-NET → OIL MASK → LOOK-ALIKE CHECK → WIND + CURRENT → OPENDRIFT/OPENOIL → ORIGIN PROBABILITY REGION → AIS → TRAJECTORIES → CORRELATION → SCORING → RANKED VESSELS → DASHBOARD → EVIDENCE REPORT

PART D — WHAT HAPPENS AT EVERY STEP

#	Step	What Happens	Data Source	Output
1	User selects area/date	User gives map area + time window.	User input	AOI + time window
2	Find Sentinel-1 scene	Search satellite catalogue for matching SAR scenes.	Copernicus Data Space	Scene metadata
3	Download SAR	Download selected Sentinel-1 GRD product.	Copernicus	GRD raster
4	Preprocess SAR	Calibrate/normalize/mask/tile the image for ML.	Sentinel-1 product	ML-ready tiles
5	Detect oil	U-Net predicts oil probability per pixel.	Trained public-data model + new SAR image	Probability mask
6	Create spill polygon	Convert mask into geographic spill boundary.	U-Net mask + georeferencing	GeoJSON polygon
7	Verify look-alike	Check wind, shape, contrast and optional classifier.	SAR + environmental data	Verified/uncertain/false
8	Get wind/current	Retrieve environmental fields around event time/location.	Copernicus Marine + chosen weather source	Wind/current fields
9	Reverse drift	Run particles backward/ensemble to infer origin.	Spill + forcing + OpenOil	Particle tracks
10	Build origin region	Aggregate particle density into a probability region.	Drift particles	Origin probability map
11	Get AIS	Find vessels present in origin region/time window.	AISStream prototype / authorized provider	AIS positions
12	Clean AIS	Remove duplicates/bad jumps and flag gaps.	Raw AIS	Clean positions
13	Build trajectory	Connect time-ordered positions into tracks.	Clean AIS	LineString trajectories
14	Find candidates	Spatial/time query for vessels compatible with origin.	PostGIS + AIS + origin region	Candidate vessels
15	Score	Calculate proximity/time/trajectory/heading/speed/AIS-quality factors.	All previous outputs	Explainable scores
16	Rank	Sort vessels by score and show reasons.	Scores	Ranked list
17	Show dashboard	Display spill, origin, ships, tracks and score breakdown.	FastAPI + PostGIS	Investigation UI
18	Generate report	Package evidence, sources, model versions and uncertainty.	All case artifacts	PDF/report

PART E — STEP-BY-STEP IMPLEMENTATION PLAN

Important rule: we first build the product with dummy data, then replace each dummy source with real data. This prevents integration problems.

Step	Implementation	Data Source	Done When
Step 0 — Git + project setup	Create repository and folders: frontend, backend, worker, ml, infra.	No external data; only code/config.	Team can clone and run the basic project.
Step 1 — Docker	Create Docker Compose for Next.js, FastAPI, PostGIS and Redis.	Local Docker images.	All services start with one command.
Step 2 — Database	Create PostgreSQL/PostGIS schema for cases, spills, vessels, AIS, drift and scores.	No external data initially; seed dummy data.	Spatial query works.
Step 3 — Backend	Build FastAPI health, case CRUD and job-status APIs.	Dummy JSON/database data.	Swagger/API docs work.
Step 4 — Frontend	Build Next.js dashboard and MapLibre map.	Dummy spill + dummy vessels.	User sees spill, ships and score on map.
Step 5 — Job system	Add Redis queue and worker so heavy tasks run in background.	Dummy jobs first.	Frontend shows queued/running/completed/failed.
Step 6 — Real Sentinel-1 search	Backend searches Copernicus by AOI/date and saves product metadata.	Copernicus Data Space Sentinel-1 catalogue.	Real scene metadata appears in the case.
Step 7 — Real Sentinel-1 download	Download a selected GRD scene and store it in S3/MinIO.	Copernicus Data Space.	Real GeoTIFF/SAFE artifact stored.
Step 8 — SAR preprocessing	Use GDAL/Rasterio/SAR tooling to produce normalized/tiled inputs.	Real Sentinel-1 GRD.	ML-ready tiles.
Step 9 — ML dataset	Download labelled oil/no-oil/look-alike dataset and create DataLoaders.	Zenodo public dataset.	Train/validation/test pipeline.
Step 10 — U-Net training	Train U-Net, measure Dice/IoU/precision/recall, save best checkpoint.	Labelled SAR dataset.	Versioned model artifact.
Step 11 — Real inference	Run saved U-Net on fresh Sentinel-1 tiles, merge mask and polygonize.	Fresh Sentinel-1 + saved model.	Real spill polygon.
Step 12 — Look-alike verification	Add transparent checks and later optional classifier.	SAR + wind + shape/contrast + look-alike dataset.	Verification confidence/status.
Step 13 — Environmental adapters	Implement wind/current data download/subsetting.	Copernicus Marine + chosen weather source.	Time/space environmental files.
Step 14 — OpenDrift/OpenOil	Run test forward simulations, then backward/ensemble hindcasting.	Spill polygon + wind/current.	Origin probability region.
Step 15 — AIS backend	Create server-side AISStream WebSocket worker and persist messages.	AISStream prototype feed.	AIS positions in PostGIS.
Step 16 — AIS cleaning + trajectories	Deduplicate, validate, flag gaps and build vessel LineStrings.	Stored AIS positions.	Clean trajectories.
Step 17 — Correlation	Query vessels near the origin region during the inferred time window.	Origin region + AIS trajectories.	Candidate list.
Step 18 — Scoring	Calculate factor scores and final rank; store all factor values.	Drift + AIS + spill features.	Explainable ranked vessels.
Step 19 — Evidence dashboard	Add timeline, layer toggles, vessel details and report generation.	All previous artifacts.	Full investigation workflow.
Step 20 — Testing	Run unit, integration, GIS, ML and end-to-end tests.	Synthetic + public labelled cases.	Validated release candidate.
Step 21 — Deployment	Deploy containers, secrets, storage, DB backup, logs and monitoring.	AWS or equivalent.	Remote working system.

PART F — FIRST VERSION / MVP

For SIH, do not wait for the entire global production system. The MVP should prove the complete reasoning chain.

- User selects a case area and time.

- A real or prepared Sentinel-1 scene is loaded.
- U-Net detects an oil-like region.
- System verifies/flags look-alikes.
- Wind/current are loaded.
- OpenOil/OpenDrift produces an origin probability region.
- AIS positions/trajectories are shown.
- At least 3 candidate vessels are ranked.
- Each score is explained factor-by-factor.
- A report is generated.
- UI clearly says 'investigative probability', not legal guilt.

PART G — DATABASE TABLES

Table	Main Fields
cases	case_id, status, AOI, start_time, end_time, created_at
satellite_scenes	product_id, acquisition_time, bbox, polarization, storage_uri
spill_detections	spill_id, case_id, confidence, area_km2, polygon, model_version
verification_results	spill_id, wind/shape/contrast_features, status, confidence
environmental_runs	run_id, source, variables, time_range, spatial_extent
drift_runs	run_id, spill_id, model_parameters, origin_geometry, confidence
vessels	vessel_id, MMSI, IMO, name, type, flag
ais_positions	vessel_id, timestamp, lat, lon, SOG, COG, geometry
trajectories	vessel_id, time_range, LineString, gap_stats
attributions	spill_id, vessel_id, all_factor_scores, final_score, rank
jobs	job_id, case_id, job_type, status, progress, error
evidence_artifacts	case_id, artifact_type, storage_uri, checksum
model_versions	name, version, metrics, artifact_uri

PART H — FRONTEND PAGES

Page	Purpose
Login	User authentication.
Case List	Previous investigations.
Create Case	Select map area and date/time.
Investigation Map	Spill + origin + vessels + trajectories.
Spill Details	Detection and verification details.
Drift View	Origin simulation/probability region.
Vessel Ranking	Rank + score + explanation.
Vessel Details	AIS trajectory and event timeline.
Evidence Report	All evidence and export.
Admin	Model/data source/system status.

PART I — BACKEND SERVICES

Service	Responsibility
Satellite Service	Search/download Sentinel-1 scenes.
Preprocessing Service	Prepare SAR tiles.
ML Service	Run U-Net inference.
Verification Service	Look-alike checks.
Environmental Service	Wind/current retrieval.
Drift Service	OpenOil/OpenDrift jobs.
AIS Service	Ingest/normalize AIS.
Trajectory Service	Build vessel tracks.
Correlation Service	Find candidate vessels.
Scoring Service	Rank and explain vessels.
Report Service	Generate evidence report.

PART J — SCORING: SIMPLE FIRST VERSION

Initial weights are engineering defaults for the prototype, not scientifically validated legal probabilities. They must be calibrated on labelled/validated cases.

Final Score = $0.35 \times \text{Origin Proximity} + 0.20 \times \text{Time Match} + 0.15 \times \text{Trajectory Match} + 0.10 \times \text{Heading Match} + 0.10 \times \text{Speed Match} + 0.10 \times \text{AIS Reliability}$

Factor	Simple meaning
Origin proximity	Ship was close to the high-probability origin area.
Time match	Ship was there during the inferred discharge window.
Trajectory match	Ship's path is consistent with entering/leaving the origin.
Heading match	Direction is compatible with movement/event.
Speed match	Speed is plausible for the event.
AIS reliability	How complete/clean the available AIS history is; gaps lower confidence.

PART K — WHAT WE WILL NOT DO

- We will not say the nearest vessel is automatically responsible.
- We will not treat an AIS gap as proof of illegal activity.
- We will not call an 88% score an 88% legal probability unless scientifically calibrated.
- We will not expose AIS/API keys in the frontend.
- We will not store huge GeoTIFF/NetCDF files directly inside PostgreSQL.
- We will not start by building a huge global real-time system.
- We will not claim public AIS coverage is complete for every Indian-water scenario.

PART L — TEAM WORK SPLIT

Member	Primary Work
Member 1	Next.js + TypeScript + MapLibre dashboard
Member 2	FastAPI + PostgreSQL/PostGIS + Redis
Member 3	Sentinel-1 preprocessing + U-Net + look-alike
Member 4	AIS + OpenDrift/OpenOil + trajectory + scoring
Everyone	Integration, testing, SIH demo, documentation

PART M — THE EXACT ORDER WE SHOULD START CODING

1. Day 1: Git repo + Docker + Next.js + FastAPI + PostGIS + Redis.
2. Day 2: Database tables + FastAPI case APIs.
3. Day 3: Map dashboard + dummy spill polygon + dummy vessels.
4. Day 4: Dummy scoring + ranked vessel UI.
5. Day 5: Redis worker + job progress.
6. Next: connect real AIS.
7. Then: connect real Sentinel-1.
8. Then: train/use U-Net.
9. Then: add wind/current.
10. Then: add OpenOil/OpenDrift.
11. Finally: integrate everything and generate evidence report.

PART N — SIMPLE MENTAL MODEL

SAR = satellite ki radar image.

U-Net = image ko dekhkar oil pixels find karne wala AI.

AIS = ship tracking data.

Wind + current = oil ko move karne wale environmental forces.

OpenOil/OpenDrift = oil movement simulation.

Reverse drift = spill se peeche jaakar possible origin.

PostGIS = location-aware database.

Scoring = saare clues ko combine karke vessels ko rank karna.

Dashboard = analyst ko saari evidence ek jagah dikhana.

PART O — PRD ACCEPTANCE CRITERIA

- A new case can be created from the UI.
- The case can find/select a Sentinel-1 scene.
- A processed SAR image can reach the ML model.
- The model returns a georeferenced spill mask/polygon.
- The system can flag/reject at least known look-alike examples.
- Environmental forcing can be retrieved for a case.
- A drift run produces a reproducible origin probability region.
- AIS data can be ingested into PostGIS.
- Candidate vessels can be filtered by time and location.
- The scoring engine produces a factor-by-factor ranking.
- The dashboard visualizes all major layers.
- The evidence report records data sources, timestamps and model versions.
- The product clearly labels attribution as investigative/probabilistic.

PART P — REFERENCES / OFFICIAL DATA DOCUMENTATION

- Copernicus Data Space — Sentinel-1 mission/data collection.
- Copernicus Data Space — Sentinel-1 GRD API documentation.
- Copernicus Marine — ocean/wind data products and access.
- AISStream — WebSocket AIS developer documentation.
- OpenDrift — official tutorial/OpenOil documentation.
- Zenodo — Sentinel-1 SAR oil-spill dataset.

END OF PRD